KRISH JOSHI

Agentic Software Engineer & Terminal SWE Benchmark Author | Trajectory Verification

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PROFESSIONAL SUMMARY

Agentic Software Engineer, **Terminal SWE Benchmark Author**, and 3rd-year CS undergraduate (Class of 2028) who builds production software and evaluation systems by directing autonomous AI coding agents (Google Antigravity, Claude Code, OpenAI Codex). Experienced in authoring Docker Harbor evaluation environments, designing rigorous test rubrics, and engineering deterministic verifiers. Author of **AgentTrace**, an open-source trajectory observatory and action-hallucination interceptor that validates agent actions against ground-truth execution logs.

CORE COMPETENCIES & TECHNICAL SKILLS

- **Agentic Orchestration & Autonomy:** Google Antigravity, Claude Code / Opus, OpenAI Codex, Model Context Protocol (MCP), Spec-Driven Workflows, Multi-Agent Loops
- **Benchmark Design & Sandboxing:** Docker Harbor / Terminal-Bench Standards, Benchmark Rubric Design, Ephemeral Sandboxes, CTRF Schemas, SHA-256 State Assertions
- **Verification & Test Engineering:** Deterministic Two-Tier PyTest Frameworks, Smoke & Deep Verifiers, Path Traversal Defense (assert_safe_path), Regression Detection
- **Trajectory Forensics & Tooling:** Trajectory Auditing (JSONL Traces), Action Hallucination Interception (Ghost Actions, Masked Errors, Circular Loops), Python, Bash/Linux, Git

PROFESSIONAL EXPERIENCE

Handshake — Frontier AI Evaluation Program **July 2026 – October 2026 (Completed)**

Terminal SWE Benchmark Author & Agent Trajectory Auditor *100% Quality Score | Remote Contractor*

- **Docker Harbor SWE Benchmarks (Project Dynamo, Jul–Sep 2026):** Authored and audited 10 terminal-based engineering environments in Docker Harbor / Terminal-Bench format to benchmark frontier coding agents on GPU kernel optimization, system recovery, and repo-scale refactoring.
- **Rubric Design & Two-Tier Verification:** Designed rigorous evaluation rubrics, two-tier PyTest verifiers (smoke probe + deep verification), and cryptographic SHA-256 state assertions to prevent agent false passes and catch subtle model bugs.
- **Trajectory Auditing & Grounding (Project Octahedron, Oct 2026):** Logged 27+ hours analyzing 32 multi-turn agent reasoning and tool-calling trajectories across frontier LLMs, diagnosing tool-call grounding errors, context window saturation, and unhandled API 401/403/429 loops.
- **Pass-Rate Calibration & Error Analysis:** Calibrated benchmark task difficulty across model families, tuning prompts and verifiers to reliably expose failure modes in multi-step planning, file editing, and test execution.

FEATURED ENGINEERING PROJECTS

AgentTrace — Forensic Agent Observatory & Action-Hallucination Interceptor **github.com/KR-007J/ai-systems-eval-portfolio**

Flagship Systems Tool | Python, Docker, PyTest, CTRF Schema, Web UI *Open Source (MIT)*

- Architected an open-source trajectory auditing engine that detects and intercepts **Action Hallucinations** (Ghost Actions, Masked 403/429 Failures, and Circular Loops) before ungrounded agent responses reach users.
- Implemented calibrated SQS (Structured Quality Score 0.0–5.0) grading rubrics, automated CTRF test reporting, and path-traversal sandboxing (assert_safe_path); verified across 22 automated test suites running in 0.03s.
- Engineered interactive terminal CLI (agenttrace audit/demo) and a local visual dashboard for inspecting step-by-step tool execution receipts and cryptographic state deltas.

Automated Docker Harbor Evaluation Engine & Recovery Probe **Python, Docker, PyTest, Harbor Standards**

- Built an automated testing harness that provisions ephemeral Docker containers, injects agent-generated multi-file patches, and executes two-tier verifications against SHA-256 state assertions.
- Developed trajectory parsing scripts to inspect multi-turn agent interaction logs, measuring tool selection accuracy, argument schema drift, and error recovery under simulated API rate limits.

EDUCATION & PROFESSIONAL CERTIFICATIONS

Bachelor of Technology (B.Tech) in Computer Science & Engineering **July 2024 – June 2028 (Expected)**

Institute of Advanced Research (IAR) — Gandhinagar, Gujarat, India **Current Status: 3rd-Year Undergraduate**

- **Google Cybersecurity Professional Certificate:** Linux security controls, asset defense, vulnerability triage, Python automation.
- **NVIDIA Introduction to Networking:** Modern network architectures, low-latency communication, distributed systems interconnects.